

Exercise 1: Number of trainable parameters: 1693041

Exercise 2: Number of trainable parameters: 15053203

Exercise 3: if efficient refers to the speed of training, then the simple feedforward neural network model is more efficient and will run faster, since it has less trainable parameters

Exercise 4: `torch.Size([3, 224, 224])`

Exercise 5: `torch.Size([3, 224, 224])`

Exercise 6: the CNN performed better than the feedforward model by about 8 percent. This is because CNNs are specifically designed for image data and can effectively capture spatial and hierarchical features, and the feedforward networks treat images as flat vectors which would be less accurate.

Exercise 7:

```
count of parameters in the model = 47.844874 million
training: 0%|                                     | 0/20 [00:00<?, ?it/s]
training loss: 2.363319158554077 -- iteration = 0
training loss: 2.1684465408325195 -- iteration = 500
training loss: 2.2058889865875244 -- iteration = 1000
training loss: 1.8872127532958984 -- iteration = 1500
training loss: 1.9482269287109375 -- iteration = 2000
training loss: 2.1781716346740723 -- iteration = 2500
training loss: 1.758748173713684 -- iteration = 3000

-----Test Accuracy = 34.35

-----saving model-----
training: 0%|                                     | 0/20 [18:22<?, ?it/s]
Traceback (most recent call last):
  File "/home/maya_geva_uri_edu/CSC592/Assignment3/VisionTransformerMG/main.py", line
157, in <module>
    sys.exit(int(main() or 0))
    ^^^^^
  File "/home/maya_geva_uri_edu/CSC592/Assignment3/VisionTransformerMG/main.py", line
150, in main
    torch.save(checkpoint_data, ckpt_path)
  File "/home/maya_geva_uri_edu/pytorch-env/lib/python3.12/site-packages/torch/seriali
zation.py", line 964, in save
    with _open_zipfile_writer(f) as opened_zipfile:
    ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/maya_geva_uri_edu/pytorch-env/lib/python3.12/site-packages/torch/seriali
zation.py", line 828, in _open_zipfile_writer
    return container(name_or_buffer)
    ^^^^^^^^^^^^^^^^^
  File "/home/maya_geva_uri_edu/pytorch-env/lib/python3.12/site-packages/torch/seriali
zation.py", line 792, in __init__
    torch._C.PyTorchFileWriter(
RuntimeError: Parent directory checkpoint does not exist.
```